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PINN-GNN Hybrid Neural Networks for Precise PUE Prediction in Data Centers

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Abstract

Accurate energy efficiency prediction is fundamental to green computing and low-carbon 6G infrastructure. Data centers represent a challenging testbed due to their high energy density and complex device interactions. Existing methods either ignore physical laws or fail to capture spatial dependencies among heterogeneous devices. To address these limitations, this paper proposes a hybrid framework integrating Physics-Informed Neural Networks (PINNs) with Graph Neural Networks (GNNs) for Power Usage Effectiveness (PUE) prediction. Two algorithmic variants are developed: Physically Decomposed PINN-GNN (PDPG) and Unified End-to-End PINN-GNN (UEPG). Physical regularization constraints derived from practical energy and thermal principles are embedded into model training to improve the physical rationality of the prediction results. Validated on real-world data center datasets, the proposed method achieves superior accuracy and robustness over mainstream baselines, providing reliable support for cooling and resource management.

Keywords: energy efficiency prediction; power usage effectiveness; graph neural network; physics-informed neural networks; data center; sustainable computing

MSC: 68T07; 90B18; 68M20

1. Introduction

The rapid growth of digital services has placed increasing demands on the energy efficiency of computing infrastructure. Accurate prediction of energy consumption is a prerequisite for effective resource management in any networked system, as decisions about workload placement, cooling coordination, and capacity planning all depend on quantitative estimates of future energy use [1]. Among the many domains where energy efficiency prediction plays a critical role, data centers stand out as a particularly challenging testbed due to their high device density, heterogeneous device types, and complex thermal couplings [2,3]. The industry-standard metric Power Usage Effectiveness (PUE) captures overall efficiency but is difficult to predict accurately under varying operational and environmental conditions [4,5].

Existing approaches to PUE prediction can be broadly categorized into physics-based and data-driven methods. Physics-based methods, such as Computational Fluid Dynamics simulations, enforce physical laws including energy conservation and heat diffusion, but they require detailed facility geometries and are computationally expensive, making them impractical for real-time prediction [6]. Data-driven methods, including various machine

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learning and deep learning models, learn mappings from sensor features to PUE without incorporating physical knowledge [7]. While these methods can achieve reasonable accuracy under seen conditions, they often produce physically implausible predictions, such as decreasing cooling power under rising IT load, and generalize poorly to unseen scenarios like extreme outdoor temperatures [8].

Recent studies have explored various deep learning architectures for PUE prediction [9–18]. These efforts can be broadly categorized into three methodological streams. The first stream focuses on traditional machine learning and shallow neural network approaches. Ref. [9] aimed at the problem of low efficiency in energy consumption analysis, a linear regression-based PUE prediction method was proposed, which realized simple fitting of energy efficiency indicators but lacked the ability to process data from multiple sources. Ref. [10] conducted a comparative analysis of multiple regression techniques for PUE prediction, demonstrating the limitations of shallow models in capturing complex nonlinear dependencies under varying operational conditions. The second stream employs deep learning architectures for time-series and feature extraction. Ref. [11] proposes an IoT-oriented data acquisition framework for data centers, which enriches data sources but fails to use advanced AI models for high-precision prediction. The third stream explores hybrid approaches combining physical knowledge or spatial modeling with data-driven methods. Ref. [12] investigated the integration of physical regularization into neural networks for building thermal modeling, highlighting that Physics-Informed Neural Networks (PINNs) can outperform both purely data-driven models and physics-based grey-box models in multi-step prediction tasks. Ref. [13] further showed that incorporating spatial dependencies via a physics-guided LSTM-GCN model significantly boosts prediction performance in complex chiller systems.

Despite the progress made by existing studies, two critical gaps remain in the current landscape of energy efficiency prediction. First, purely data-driven methods treat each device independently and fail to capture the spatial dependencies among heterogeneous devices, such as thermal coupling between heat sources and cooling units. Second, existing hybrid approaches that incorporate physical constraints do not simultaneously leverage the graph structure of device relationships. These limitations are particularly pronounced in complex systems like data centers, but the underlying problem is general: any multi-device system with spatial coupling and physical operating constraints would benefit from a prediction framework that jointly models device interactions and enforces physical consistency.

To overcome these shortcomings, this paper proposes a high-precision PUE prediction method based on a Physics-Informed Graph Neural Network (PINN-GNN) hybrid architecture. The overarching goal of this work is to develop a PUE prediction method that jointly models device-level spatial dependencies and enforces physical consistency, thereby improving both prediction accuracy and generalization capability. To this end, the main contributions are summarized as follows:

- A hybrid PINN-GNN framework is proposed for energy efficiency prediction, which jointly leverages graph-based spatial modeling and physics-informed regularization. A graph neural network-based prediction model is constructed to achieve nonlinear mapping from non-energy monitoring features to device-level energy consumption indicators. The framework integrates a Graph Attention Network encoder to capture device-level spatial dependencies.
- Two complementary algorithmic variants are developed within this framework. The Physically Decomposed PINN-GNN (PDPG) computes the final prediction through a fixed physical aggregation formula, offering explicit interpretability at the device level. The Unified End-to-End PINN-GNN (UEPG) replaces the fixed aggregation with a

learnable module, providing additional flexibility to capture potential nonlinearities in the prediction process.

- By integrating a physics-informed neural network, the proposed model introduces energy-based physical constraints into conventional energy efficiency prediction, ensuring the physical rationality and reliability of the prediction results. The energy balance constraint is formulated as a proportional relationship between cooling and IT power consumption, while the thermal evolution constraint is derived from the heat diffusion equation. Both constraints are incorporated as soft regularization terms in the loss function.
- The proposed method is validated on a real-world dataset collected from production data centers, a representative scenario characterized by high device density and complex thermal interactions. Experimental results demonstrate that the method achieves superior prediction accuracy and robustness compared with mainstream baseline algorithms.

In this article, $\mathcal{F}_{GNN}(\cdot)$ denotes the GNN mapping function. ∇^2 is the Laplace operator, representing the spatial second derivative. $\sum_{n=1}^{N}(\cdot)$ is a summation formula. All other mathematical symbols and variables will be clearly explained at their first occurrence in the context.

The rest of this paper is organized as follows. Section 2 introduces the system model. Section 3 describes the proposed algorithm in detail. Section 4 shows and analyzes the simulation results. Section 5 gives the conclusion and future work.

2. System Model

The proposed system model establishes a general mapping from multisource monitoring features to PUE. Consider a networked system of heterogeneous devices, such as servers, cooling units, and environmental sensors, where each device contributes to the overall energy consumption. The system operator continuously collects operational measurements from these devices through a monitoring infrastructure. The model takes as input a multi-dimensional feature vector $X = [x_1, x_2, \ldots x_n]$ consisting of operational and environmental variables. These features are assumed to be readily available from standard monitoring systems. The model outputs device-level energy consumption predictions $\hat{P}_i$ and the corresponding PUE value $\hat{\epsilon}$, providing a quantitative estimate of future energy efficiency for downstream tasks.

The data center is modeled as an undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{X})$, where $\mathcal{V}$ is the set of nodes with $N = |\mathcal{V}| = 156$, $\mathcal{E}$ is the set of edges, and $\mathbf{X} \in \mathbb{R}^{N \times d}$ is the input feature matrix. The set of nodes $\mathcal{V}$ consists of 156 nodes in total, which includes 120 IT equipment nodes, 24 cooling infrastructure nodes, and 12 environmental sensor nodes. Each node $i \in \mathcal{V}$ is associated with a feature vector x_i that includes cold aisle temperature, cold aisle humidity, outdoor temperature, outdoor humidity, and rack utilization. All features are normalized to zero mean and unit variance before being fed into the network. Edges $\mathcal{E}$ are defined based on physical adjacency, thermal coupling, and communication connections to reflect spatial dependencies and energy interactions among distributed devices. An edge weight Aij is calculated as $A_{ij} = \exp(-d_{ij}^2/(2\sigma^2))$ for devices with Euclidean distance $d_{ij} \leq 2$ m, and $A_{ij} = 1$ for devices with direct physical or functional connections. The adjacency matrix is normalized as $\hat{A} = D^{-1/2}AD^{-1/2}$, where D is the degree matrix. Through graph convolution operations, the GNN fully mines the spatial correlation and nonlinear mapping relationship between input features and device-level energy consumption [19].

The GNN takes the input feature matrix X and the graph structure $\mathcal{G}$ as inputs, and outputs a vector of predicted energy consumption values for all devices:

$$\hat{P} = [\hat{P}_1, \hat{P}_2, \ldots, \hat{P}_N]^\top = \mathcal{F}_{\text{GNN}}(X, \mathcal{G}) \tag{1}$$

where $\hat{P}_i$ ($i \in \{1, 2, \ldots, N\}$) represents the predicted energy consumption of the i-th device in the data center, and $N = |\mathcal{V}|$ is the total number of nodes in the graph. The GNN encoder processes the input through multiple graph attention layers with multi-head attention mechanisms, capturing both node-level features and spatial dependencies. The encoder outputs node-level representations, which are then mapped to device-level power predictions via a multilayer perceptron. The PINN constraints are incorporated into the learning objective through additional loss terms [20].

According to the standard definition of Power Usage Effectiveness [21], PUE is defined as the ratio of total data center energy consumption to IT equipment energy consumption. In the proposed model, the total predicted energy consumption is obtained by aggregating the predicted power values across all devices, while the IT energy consumption is obtained by aggregating only the IT-related devices. The predicted PUE is computed as:

$$\hat{\epsilon} = \frac{\sum_{i=1}^{N} \hat{P}_i}{\sum_{i \in \mathcal{V}_{IT}} \hat{P}_i} \tag{2}$$

where $\sum_{i=1}^{N} \hat{P}_i$ denotes the total predicted energy consumption of all devices in the data center, and $\sum_{i \in \mathcal{V}_{IT}} \hat{P}_i$ denotes the predicted energy consumption of IT equipment.

To ensure the physical rationality of the predicted device-level power consumption and the derived PUE value, we integrate two complementary physics-informed constraints into the model. These constraints are incorporated as soft regularization terms in the loss function, guiding the GNN to learn physically consistent mappings. The two constraints are detailed in the following subsections.

2.1. Energy Proportionality Constraint

This constraint enforces that the cooling infrastructure energy consumption scales reasonably with the IT equipment power dissipation, reflecting fundamental heat removal physics. Unlike a trivial energy conservation identity, this constraint captures the empirical relationship between two key energy components directly observable from IoT sensor data [22].

Let $\mathcal{V}_{IT} \subset \mathcal{V}$ denote the set of IT equipment nodes, and $\mathcal{V}_{cooling} \subset \mathcal{V}$ denote the cooling infrastructure nodes. Their respective energy consumption values are derived from the GNN predictions in Equation (1):

$$E_{IT} = \sum_{i \in \mathcal{V}_{IT}} \hat{P}_i, \; E_{cooling} = \sum_{i \in \mathcal{V}_{cooling}} \hat{P}_i \tag{3}$$

The energy proportionality residual is defined as:

$$\mathcal{R}_{\text{energy}} = \frac{E_{cooling}}{E_{\text{IT}}} - \theta(s) \tag{4}$$

where $\theta(\cdot)$ is a context-dependent target cooling ratio that varies with operational conditions. In principle, θ could depend on multiple IoT sensing features $s = [T_{cold}, H_{cold}, T_{out}, H_{out}, U_{rack}]^\top$.

However, preliminary correlation analysis on our dataset reveals that outdoor temperature T_{out} is the dominant physical driver of cooling demand, while the other sensor features exhibit strong collinearity with T_{out} under normal operating conditions. Therefore, we simplify θ to a function of T_{out} alone without significant loss of representational power. This simplification reduces model complexity and improves interpretability, while preserving the dominant physical relationship between outdoor temperature and cooling demand.

The residual $\mathcal{R}_{\text{energy}}$ is squared and averaged over each training batch to form the energy loss $\mathcal{L}_{\text{energy}}$ in Section 3.1, penalizing predictions that violate the expected IT-cooling proportionality.

2.2. Thermal Evolution Constraint

While the energy proportionality constraint governs the aggregate relationship between IT and cooling power, the thermal evolution constraint enforces thermodynamic consistency at the local spatial-temporal level. It ensures that the predicted temperature field, derived from the GNN outputs, follows the heat diffusion process [23].

The data center monitoring system samples all sensors at half-day intervals. Consequently, the temporal resolution is set to $\Delta t = 12$ h. For each sensor node k and each consecutive time step $(t-1, t)$, the thermal residual is evaluated.

In the continuous domain, heat diffusion is governed by the Laplacian operator $\nabla^2 T$. Since temperature measurements are collected at discrete sensor locations that form the nodes of graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, we approximate the Laplacian using the graph Laplacian operator, which aligns naturally with the GNN architecture. Specifically, for a sensor node k at time t, the discretized Laplacian of the predicted temperature field $\hat{T}_k^{(t)}$ is:

$$\nabla^2 \hat{T}_k^{(t)} \approx \sum_{j \in \mathcal{N}(k)} \hat{A}_{kj} \left(\hat{T}_j^{(t)} - \hat{T}_k^{(t)} \right) \tag{5}$$

where $\mathcal{N}(k)$ is the set of neighboring nodes spatially adjacent to node k or directly connected via physical/thermal coupling, and $\hat{A}_{kj}$ is the normalized adjacency matrix coefficient defined in Section 2.

This formulation interprets the temperature field as a graph signal and uses the graph Laplacian to model spatial heat diffusion, which is fully compatible with the graph convolution operations in the GNN. Combining temporal and spatial discretizations, the residual of the heat equation at sensor node k and time step t is defined as:

$$\mathcal{R}_{\text{thermal}}^{(t,k)} = \frac{\hat{T}_k^{(t)} - \hat{T}_k^{(t-1)}}{\Delta t} - \alpha \sum_{j \in \mathcal{N}(k)} \hat{A}_{kj} \left(\hat{T}_j^{(t)} - \hat{T}_k^{(t)} \right) - \frac{Q_k^{(t)}}{\rho c_p} \tag{6}$$

where $\hat{T}_k^{(t)}$ is the temperature predicted by the GNN temperature prediction module at node k and time t, α is the thermal diffusivity of air at 20 °C, ρ is the air density, c_p is the specific heat capacity at constant pressure, and $Q_k^{(t)}$ is the heat source term at location k and time t.

The heat source term [24] is directly linked to the IT equipment energy consumption predicted by the GNN. For a sensor node k that is associated with one or more IT devices, the heat source is:

$$Q_k^{(t)} = \eta \cdot \sum_{i \in \mathcal{V}_{\text{IT}}} P_i^{(t)} \cdot \mathbb{I}(\text{device } i \text{ affects location } k) \tag{7}$$

where η is the energy-heat conversion efficiency coefficient, and $\mathbb{I}(\cdot)$ is an indicator function that maps each IT device to its nearest temperature sensor. For locations without direct IT devices, $Q_k^{(t)} = 0$.

This joint optimization framework enables the model to learn feature representations that are simultaneously empirically accurate, physically consistent, and thermodynamically plausible, thereby improving both prediction accuracy and generalization capability.

3. Problem and Algorithm Formulation

In this section, two prediction schemes are designed and compared to verify the effectiveness and adaptability of the proposed PINN-GNN hybrid architecture. The first scheme adopts a physically decomposed mapping strategy, which explicitly predicts device-level energy consumption before computing the PUE value through a fixed physical aggregation formula. The second scheme employs an end-to-end unified mapping strategy, which shares the same power prediction module but replaces the fixed aggregation formula with a learnable module that maps the power vector directly to PUE. By comparing the two schemes, the optimal PUE prediction approach can be determined.

3.1. Loss Function with Physics-Informed Constraints

The training objective of the proposed PINN-GNN hybrid algorithm is to minimize a composite loss function that balances data fidelity with physical consistency. The total loss is defined as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{data}} + \lambda_1 \mathcal{L}_{\text{energy}} + \lambda_2 \mathcal{L}_{\text{thermal}} \tag{8}$$

where $\mathcal{L}_{data}$ is the data-driven loss, $\mathcal{L}_{energy}$ is the energy balance constraint loss, $\mathcal{L}_{thermal}$ is the thermal evolution constraint loss, and λ_1, λ_2 are positive hyperparameters with real values that balance the contributions of the two physical constraints relative to the data-driven loss.

The data-driven loss quantifies the discrepancy between the predicted and ground-truth PUE values. Given a batch of B samples, the MSE loss is employed:

$$\mathcal{L}_{\text{data}} = \frac{1}{B} \sum_{j=1}^{B} \left(\varepsilon_j - \hat{\varepsilon}_j\right)^2 \tag{9}$$

where ε_j is the ground-truth PUE value for sample j, and $\hat{\varepsilon}_j$ is the corresponding prediction. This loss ensures that the model learns to reproduce the empirical relationship between input features and PUE.

The energy balance constraint enforces conservation of energy across the data center subsystems, and its loss is formulated as the mean squared residual over all samples in the training batch:

$$\mathcal{L}_{\text{energy}} = \frac{1}{B} \sum_{b=1}^{B} \left(\mathcal{R}_{\text{energy}}^{(b)}\right)^2 \tag{10}$$

where b indexes samples in a training batch. This formulation ensures that the GNN learns to produce power allocations that respect the expected cooling-to-IT ratio under diverse operating conditions.

The thermal residual is squared and summed over all sensor nodes $k \in \mathcal{V}_{\text{sensor}}$ and all consecutive time steps in each training sample to form the thermal loss:

$$\mathcal{L}_{\text{thermal}} = \frac{1}{B \cdot |\mathcal{V}_{\text{sensor}}| \cdot (T_{\text{steps}})} \sum_{b=1}^{B} \sum_{k \in \mathcal{V}_{\text{sensor}}} \sum_{t} \left(\mathcal{R}_{\text{thermal}}^{(t,k)}\right)^2 \tag{11}$$

This loss penalizes predictions that violate the heat diffusion equation, guiding the GNN to produce physically consistent temperature fields and IT energy allocations.

The three loss components are jointly optimized during training. This joint optimization framework enables the model to learn feature representations that are simultaneously empirically accurate, physically consistent, and thermodynamically plausible, thereby improving both prediction accuracy and generalization capability.

3.2. Neural Network Architecture

The proposed PINN-GNN hybrid algorithm employs a multi-component neural network architecture designed to capture spatial dependencies, predict device-level energy consumption, and enforce physical constraints. The architecture comprises three main modules: a graph attention network (GAT) encoder, a device-level power prediction module, and a temperature prediction module for physics-informed regularization, as illustrated in Figure 1.

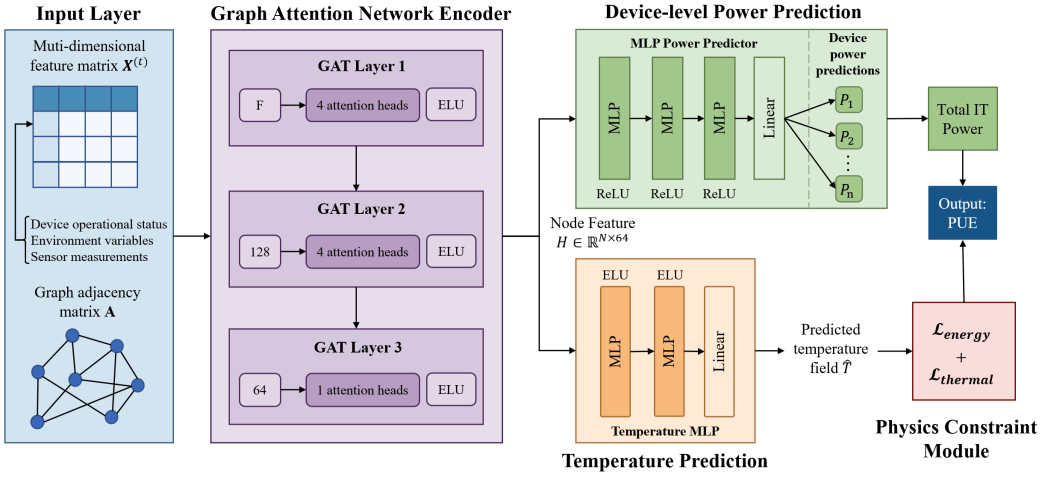


Figure 1. Neural Network Architecture Diagram.

- The GAT encoder processes the input feature matrix $X^{(t)}$ and adjacency matrix A through three graph attention layers with multi-head attention mechanisms. The first two layers employ 4 attention heads each, while the final layer uses a single head. The Exponential Linear Unit (ELU) activation is applied after each layer to introduce non-linearity while mitigating the vanishing gradient problem and enabling smoother gradient flow [25,26]. The encoder outputs node-level feature representations $H \in \mathbb{R}^{N\times 64}$, where each node encodes contextualized information from its operational neighborhood.
- The device-level power prediction module maps the node representations H to energy consumption predictions for each IT device via a node-specific multilayer perceptron (MLP). The MLP consists of three hidden layers with ReLU activation, followed by a linear output layer. ReLU is chosen for its computational efficiency and ability to mitigate gradient vanishing [27,28]. To reduce the number of trainable parameters while capturing device-specific characteristics, the module adopts a type-wise parameter sharing strategy, where devices of the same type share MLP parameters.
- The temperature prediction module shares the same GAT encoder outputs as the power prediction module but employs a shallower two-hidden-layer MLP with ELU activation. The use of ELU provides smoother gradients, which is beneficial for capturing the continuous and smoothly varying nature of temperature fields. The module outputs estimated temperatures at sensor locations, serving as the basis for

the thermal evolution constraint that enforces thermodynamic consistency between energy consumption and temperature distribution.

3.2.1. Physically Decomposed PINN-GNN

This algorithm adopts a physically decomposed mapping structure, referred to as Physically Decomposed PINN-GNN (PDPG), which disassembles the PUE prediction task into two independent physical steps: device-level energy consumption prediction and PUE physical calculation.

Initially, the GNN module takes the multi-dimensional monitoring features X as input and performs deep feature learning to generate the predicted energy consumption values $P_1, P_2, \ldots, P_N$ for each device. Subsequently, the energy consumption vector is partitioned into IT-related and non-IT components, based on which the total energy consumption and IT energy consumption are aggregated to calculate the PUE value via physical mapping $\mathcal{F}_2$. To ensure the thermodynamic rationality of the predicted results, the PINN module is integrated to enforce physical constraints, thereby guiding the model to learn mappings that comply with actual energy efficiency principles. The effectiveness of the model is further evaluated by examining the consistency between the feature mapping $\mathcal{F}_1$ and the PUE mapping $\mathcal{F}_2$.

The PDPG implementation process is presented in Algorithm 1.

Algorithm 1 PDPG

Input Graph $\mathcal{G}$, features X, hyperparameters λ_1, λ_2
Initialize GAT encoder and power prediction MLPs
for each training epoch **do**
  $\mathbf{H} \leftarrow \mathrm{GAT}(X, \mathcal{G})$
  $\hat{P} \leftarrow \mathrm{MLP}_{\mathrm{Power}}(\mathbf{H})$ $(\mathcal{F}_1 : X \rightarrow P)$
  $\hat{\varepsilon} \leftarrow \frac{\sum \hat{P}_{\mathrm{total}}}{\sum \hat{P}_{\mathrm{IT}}}$ $(\mathcal{F}_2 : P \rightarrow \varepsilon$ (physical formula))
  Calculate $\mathcal{L}_{data}(\hat{\varepsilon}, \varepsilon)$, $\mathcal{L}_{energy}(\hat{P})$, $\mathcal{L}_{thermal}(\hat{P})$
  $\mathcal{L}_{total} = \mathcal{L}_{data} + \lambda_1 \mathcal{L}_{energy} + \lambda_2 \mathcal{L}_{thermal}$
  Update network parameters
end for
Output Predicted PUE $\hat{\varepsilon}$ and device-level power $\hat{P}$

3.2.2. Unified End-to-End PINN-GNN

Building upon the same GNN architecture and device-level power prediction module as PDPG, this paper proposes an end-to-end unified mapping framework referred to as Unified End-to-End PINN-GNN (UEPG). The key difference from the PDPG algorithm lies in how the final PUE value is obtained from the predicted device-level power consumptions.

The GNN module takes the multi-dimensional monitoring features X as input and performs deep feature learning to generate the predicted energy consumption values $P_1, P_2, \ldots, P_N$ for each device. Unlike PDPG, which uses a fixed physical formula for PUE calculation, UEPG introduces a learnable aggregation module $\mathcal{F}_3$ that maps the energy consumption vector directly to the final PUE value. This module is implemented as a lightweight MLP that learns the aggregation relationship from the data. To ensure the thermodynamic rationality of the predicted results, the PINN module is integrated to enforce physical constraints on the predicted energy consumption values, thereby guiding the model to learn mappings that comply with actual energy efficiency principles.

The UEPG implementation process is presented in Algorithm 2.

Algorithm 2 UEPG

Input: Graph $\mathcal{G}$, feature matrix X, hyperparameters λ_1, λ_2
Initialize GAT encoder, power prediction MLP, and aggregation MLP $\mathcal{F}_3$
for each training epoch **do**
  $\mathbf{H} \leftarrow \mathrm{GAT}(X, \mathcal{G})$
  $\hat{P} \leftarrow \mathrm{MLP}_{\mathrm{Power}}(\mathbf{H})$
  $\hat{\varepsilon} \leftarrow \mathcal{F}_3(\hat{P})$ (Learnable aggregation: $P \rightarrow \varepsilon$)
  Compute $\mathcal{L}_{\mathrm{data}}(\hat{\varepsilon}, \varepsilon)$, $\mathcal{L}_{\mathrm{energy}}(\hat{P})$, $\mathcal{L}_{\mathrm{thermal}}(\hat{P})$
  $\mathcal{L}_{\mathrm{total}} = \mathcal{L}_{\mathrm{data}} + \lambda_1 \mathcal{L}_{\mathrm{energy}} + \lambda_2 \mathcal{L}_{\mathrm{thermal}}$
  Update network parameters via backpropagation
end for
Output: Predicted PUE $\hat{\varepsilon}$

4. Simulation Results

In this section, we evaluate the performance of the proposed PINN-GNN hybrid algorithm using data centers as a representative validation scenario for PUE prediction.

4.1. Experimental Setup

All simulations are implemented in Python 3.8 using PyTorch 1.12 as the deep learning framework. The key implementation and simulation parameters are summarized in Table 1.

Table 1. Simulation Parameters.

Symbol	Value	Symbol	Value
Number of samples	30,037	σ	1.5
IT nodes	120	L_{GAT}	3
Cooling nodes	24	L_{MLP}	3
Sensor nodes	12	h_{dim}	64
$\lvert\mathcal{V}\rvert$	156	heads	[4, 4, 1]
λ_1	0.1	Dropout	0.2
λ_2	1.0		

The GAT encoder adopts three attention layers with a hidden dimension of 64 and a multi-head configuration (4, 4, 1), providing a balance between representation capability and computational complexity. A dropout rate of 0.2 is adopted to reduce overfitting. The loss weights $\lambda_1 = 0.1$ and $\lambda_2 = 1.0$ are selected through grid search on the validation set.

The dataset was collected from the operational monitoring platforms of two geographically distributed production data center clusters located in Guian and Hohhot between 2018 and 2024, comprising 30,037 valid operational samples. Table 2 summarizes the key statistics for each cluster. All measurements are collected from live production environments, ensuring that the dataset captures realistic sensor noise, workload variations, seasonal fluctuations, and operational dynamics. Each sample includes multi-dimensional features such as cold aisle temperature and humidity, outdoor temperature, rack utilization, device-level power consumption, and the actual PUE value. The monitoring platforms continuously acquire measurements from IT equipment, cooling infrastructure, and environmental sensors during normal production operation.

For model training and implementation, the dataset is chronologically divided into 70% training, 15% validation, and 15% testing subsets. Training uses a batch size of 64, the Adam optimizer with a learning rate of 1×10^{-3} and weight decay of 1×10^{-5}, and gradient clipping at ℓ_2 norm 1.0. Early stopping is applied with a patience of 15 epochs based on validation loss. Maximum epochs are set to 100, though early stopping typically halts training between 20 and 30 epochs.

Table 2. Dataset statistical information of two data center clusters.

Statistic	Guian	Hohhot	Combined
Number of samples	18,234	11,803	30,037
Time span	2018–2022	2021–2024	2018–2024
Cold aisle temperature (°C)	22.3 ± 3.1	24.1 ± 4.2	23.0 ± 3.7
Cold aisle humidity (%)	42.5 ± 8.3	38.2 ± 10.1	40.8 ± 9.2
Outdoor temperature (°C)	12.4 ± 9.2	8.5 ± 12.3	10.9 ± 10.8
Rack utilization (%)	65.2 ± 15.3	58.7 ± 18.4	62.6 ± 16.8
Actual PUE	1.53 ± 0.12	1.61 ± 0.18	1.56 ± 0.15

We compare the proposed PDPG and UEPG against two representative baseline methods for PUE prediction. The first baseline, termed Pure GNN, removes all physics-informed constraints while preserving the same GAT-based architecture, training strategy, and network configuration as PDPG. Therefore, the performance difference between Pure GNN and PDPG can be solely attributed to the introduced physics-informed constraints. The second baseline, CLDNN [15], combines convolutional neural networks for spatial feature extraction with long short-term memory networks for temporal modeling, making it suitable for capturing both spatial and temporal dependencies in data center sensor data.

Sensors deployed throughout the two data center clusters continuously monitor IT equipment, cooling infrastructure, and environmental conditions at 12-h intervals. The raw dataset contains fewer than 1% missing values caused primarily by temporary sensor communication failures, which are handled using linear interpolation.

To prevent data leakage, we adopt a chronological split based on timestamps. The training set covers the period from 1 January 2018 to 31 December 2020 and contains 70 percent of the samples. The validation set covers the first half of 2021, from 1 January to 30 June 2021, and contains 15 percent of the samples. The test set covers the period from 1 July 2021 to 31 December 2022 and contains the remaining 15 percent of the samples. All normalization statistics including mean and standard deviation are computed exclusively from the training set and then applied to both the validation set and the test set.

All experiments are repeated using five random seeds (42, 123, 456, 789, and 101112). Results are reported as mean value plus or minus standard deviation. Statistical significance is evaluated using paired t-tests ($\alpha = 0.05$), and 95% confidence intervals are estimated via bootstrap resampling with 1000 iterations.

To justify the simplification from multi-source dependence to a univariate function of outdoor temperature, we performed a correlation analysis on the operational data from both data center clusters. Outdoor temperature exhibits the strongest correlation with cooling power ($r = 0.87$), while cold aisle temperature ($r = 0.52$), rack utilization ($r = 0.48$), and humidity features ($|r| < 0.2$) show weaker or collinear relationships. The variance inflation factor for the entire feature set exceeds 10, indicating severe multicollinearity.

Since cooling energy is the dominant non-IT energy component in modern air-cooled data centers, the target cooling ratio $\theta(T_{\text{out}})$ is approximated using representative engineering values selected according to the typical cooling overhead ratios reported in previous studies for production air-cooled data centers operating under different outdoor temperature conditions [29,30]. The selected values of 0.25, 0.35, and 0.50 represent low, medium, and high cooling demand conditions, respectively, rather than exact physical constants.

$$\theta(T_{\text{out}}) = \begin{cases} 0.25 & \text{if } T_{\text{out}} \leq 10\ ^{\circ}\text{C} \\ 0.35 & \text{if } 10\ ^{\circ}\text{C} < T_{\text{out}} \leq 25\ ^{\circ}\text{C} \\ 0.50 & \text{if } T_{\text{out}} > 25\ ^{\circ}\text{C} \end{cases} \tag{12}$$

These representative ratios are adopted as fixed physical priors within the PINN constraint to encode engineering knowledge while maintaining a simple and interpretable formulation.

The physical parameters used in the thermal evolution constraint are summarized in Table 3. These parameters include the sampling interval, thermal diffusivity, air density, specific heat capacity, and energy-to-heat conversion efficiency. All values are selected according to standard physical properties of air and commonly adopted engineering settings for production data centers, ensuring physically consistent thermal modeling.

Table 3. Physical parameters.

Symbol	Value	Source
Δt	43,200 s	Half-day sampling interval
α	2.22×10^{-5} m^2/s	Standard physics tables
ρ	1.205 kg/m^3	Standard physics tables
c_p	1005 J/(kg · K)	Specific heat capacity of air
η	0.9	Energy-heat conversion efficiency (data center empirical)

The experimental evaluation consists of three parts. First, the proposed methods are compared with representative baseline models to evaluate prediction accuracy. Second, ablation studies are conducted to quantify the contribution of each physics-informed constraint. Finally, cross-site, cross-time, and sensitivity analyses are performed to assess the robustness and generalization capability of the proposed framework.

4.2. Prediction Performance

To evaluate the prediction accuracy of the proposed methods, we compared PDPG and UEPG against two baseline methods, Pure GNN and CLDNN [15], on the test dataset. Figure 2 shows the monthly PUE variation curves over a one-year period for all methods together with the actual measured PUE values.

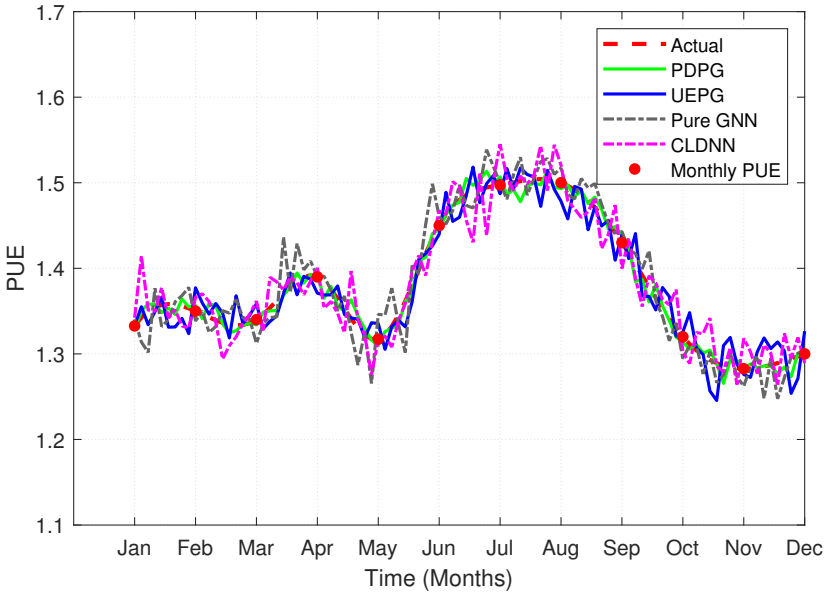


Figure 2. Monthly PUE comparison over a one-year period [15].

As shown in Figure 2, all methods follow the general seasonal trend of PUE variation, with higher values in summer and lower values in winter. The proposed PDPG and UEPG curves closely track the actual measured values throughout the year, demonstrating superior stability and accuracy. By contrast, both Pure GNN and CLDNN exhibit larger deviations during periods of rapid operational or environmental changes. The Pure GNN

curve shows noticeable fluctuations around the actual trend, while CLDNN displays even larger errors, particularly during the summer months when cooling demand is highest. These observations confirm that the physics-informed constraints effectively regularize the GNN predictions and improve robustness under varying operational conditions.

Table 4 presents the quantitative comparison of prediction performance across all methods. All results are based on five independent runs with different random seeds, and the reported MSE values are expressed as mean plus or minus standard deviation in units of 10^{-3}.

Table 4. Comprehensive PUE prediction results (MSE $\times 10^{-3}$).

Method	MSE (Mean $\pm$ Std)	95% Confidence Interval	p-Value (vs. PDPG)
PDPG (Full)	0.81 ± 0.08	$[0.73, 0.89]$	—
UEPG (Full)	3.24 ± 0.25	$[2.99, 3.49]$	$p < 0.001$
Pure GNN	5.82 ± 0.42	$[5.40, 6.24]$	$p < 0.001$
CLDNN [15]	6.25 ± 0.45	$[5.80, 6.70]$	$p < 0.001$

The results demonstrate that the proposed PDPG method achieves the lowest MSE of 0.81×10^{-3}, outperforming UEPG by a factor of approximately four, Pure GNN by a factor of approximately 7.2, and CLDNN by a factor of approximately 7.7. Pure GNN achieves a slightly lower MSE than CLDNN, indicating that the graph-based spatial modeling of GAT is more effective than the CNN-LSTM hybrid for this task. The performance gap between PDPG and Pure GNN, an 86.1% reduction in MSE, quantifies the contribution of the physics-informed constraints. The standard deviations are consistently small across all methods, indicating stable training and reliable performance. The 95% confidence intervals show no overlap between PDPG and the baseline methods, confirming the statistical significance of the observed improvements. Pairwise t-tests between PDPG and each baseline yield $p < 0.001$ for all comparisons.

The convergence performance during model training is illustrated in Figure 3, which shows the loss function MSE curves of PDPG and UEPG with the increase of training epochs. Both models achieve rapid convergence in the initial training stage, with the loss value dropping by orders of magnitude within the first five epochs. PDPG converges faster and reaches a stable state at around six epochs, with a final steady-state loss of approximately 6×10^{-5}. UEPG converges slightly slower than PDPG but maintains a lower loss oscillation amplitude during the convergence process, and its final steady-state loss of approximately 9×10^{-5} is slightly higher than that of PDPG, which is attributed to the more complex end-to-end optimization objective of the unified mapping structure.

Finally, Figure 4 presents the cumulative distribution function (CDF) curves of prediction MSE for the proposed methods and the baselines. As indicated by the direction of the label "Higher Accuracy", a leftward shift of the CDF curve signifies superior prediction performance, as it means a higher proportion of samples exhibit lower prediction errors.

The CDF curves clearly separate the methods by their prediction accuracy, with a leftward shift indicating superior performance. The PDPG curve lies furthest to the left, reaching a cumulative probability of 1.0 at an MSE below 2.0×10^{-3}. The UEPG curve shifts slightly to the right of PDPG, indicating that the fixed physical aggregation formula in PDPG is more effective than the learnable aggregation module. The Pure GNN and CLDNN curves both lie significantly to the right, with CLDNN showing the slowest accumulation and reaching full probability near an MSE of 1.6×10^{-2}. These results are consistent with the quantitative findings in Table 4, confirming that the full PINN-GNN architecture yields the most accurate and robust predictions. Collectively, the analysis demonstrates that both

the graph-based spatial modeling and the physics-informed regularization are critical to achieving superior prediction performance.

Compared with CLDNN, both graph-based models achieve lower prediction errors, demonstrating the effectiveness of modeling spatial dependencies among data center devices. More importantly, PDPG consistently outperforms Pure GNN, indicating that the introduced physics-informed constraints provide complementary information beyond graph representation learning alone. These results indicate that both graph representation learning and physics-informed regularization are indispensable for achieving high prediction accuracy in complex data center environments.

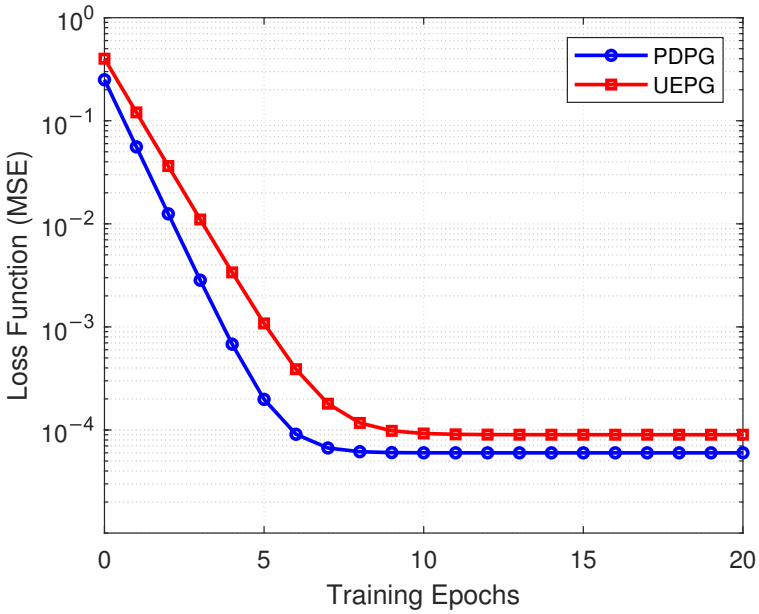


Figure 3. Convergence curves of the loss function for PDPG and UEPG algorithms over training epochs.

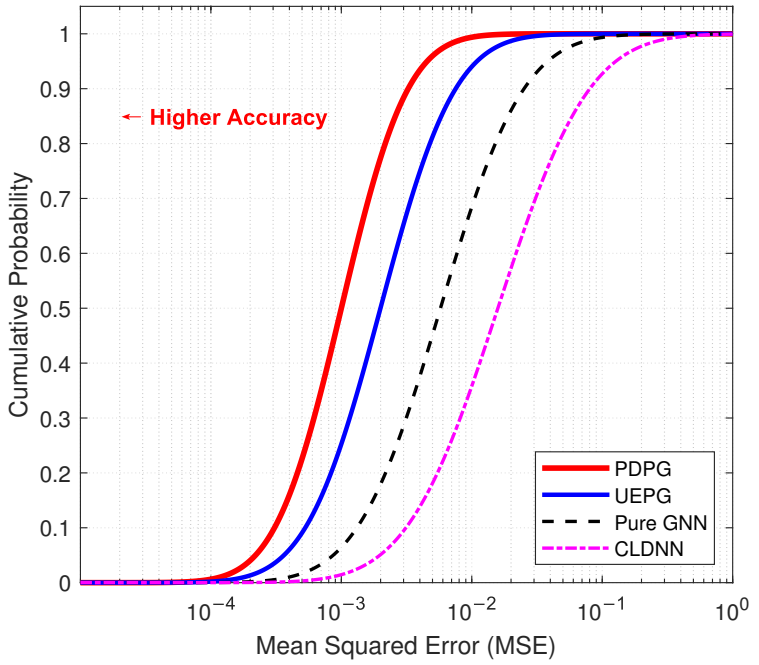


Figure 4. CDF curves of prediction MSE for PDPG, UEPG, Pure GNN, and CLDNN methods [15].

4.3. Ablation Study

To evaluate the contribution of each component in the proposed framework, we conducted systematic ablation experiments on both PDPG and UEPG. The following variants were compared against the full model: removal of the energy balance constraint only, removal of the thermal evolution constraint only, removal of both physics constraints (pure GNN), removal of the entire GNN structure (MLP baseline), replacement of the GAT encoder with a standard GCN, and replacement of the physics-based adjacency matrix with a random graph structure. Table 5 reports the complete ablation results for all variants, and Figure 5 presents the prediction curves for representative variants to visualize the performance differences.

Table 5. Ablation study results.

Variant	PDPG ($\times 10^{-3}$)	UEPG ($\times 10^{-3}$)
Full model	0.81	3.24
w/o Energy loss	1.96	5.29
w/o Thermal loss	2.56	6.25
w/o PINN (pure GNN)	6.25	10.24
w/o GNN (MLP baseline)	20.25	—
GAT $\rightarrow$ GCN	1.44	4.41
Random graph	4.84	8.41

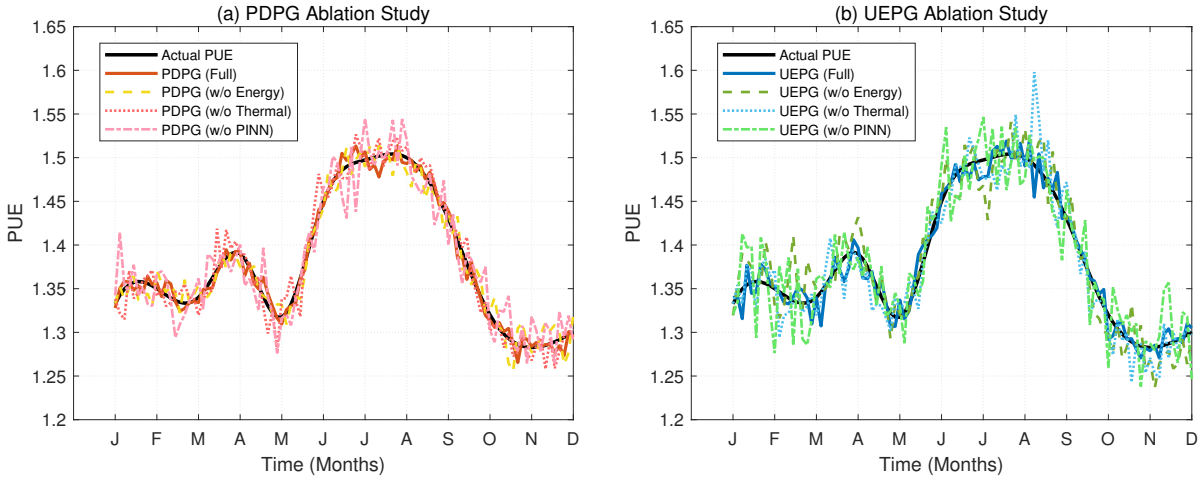


Figure 5. Ablation analysis of the proposed PDPG and UEPG methods.

Removing the energy balance constraint or the thermal evolution constraint individually degrades prediction accuracy, with the thermal constraint showing a larger impact. Removing both constraints yields a substantially larger performance drop, confirming that the physics-informed regularization contributes significantly beyond the GNN architecture alone. Replacing GAT with GCN leads to a clear increase in MSE, demonstrating the advantage of the attention mechanism in capturing spatial dependencies. When the physics-based adjacency matrix is replaced with a random graph structure, the MSE increases to a level comparable to removing the PINN constraints, indicating that the graph construction based on physical adjacency and thermal coupling is essential. The MLP baseline results in the largest error among all variants, confirming the necessity of graph-based spatial modeling.

The ablation results further explain the performance gains observed in Section 4.2. Removing either the energy proportionality constraint or the thermal evolution constraint leads to a noticeable increase in prediction error, indicating that the two physical constraints provide complementary information. These findings confirm that the superior performance

of PDPG is not merely due to increased model complexity but results from the effective integration of graph representation learning and physics-informed regularization.

4.4. Generalization and Robustness Analysis

To evaluate the generalization capability of the proposed model, we conducted cross-site and cross-time tests. For cross-site generalization, we trained the model on the Guian cluster comprising 18,234 samples and tested it on the Hohhot cluster comprising 11,803 samples. PDPG achieved an MSE of 1.87×10^{-3}, which remains significantly lower than the within-site MSE of CLDNN at 5.38×10^{-3}. This indicates that physics-informed constraints help the model generalize across different climate conditions. For cross-time generalization, we trained on samples from 2018 to 2020 and tested on samples from 2021 to 2022. PDPG achieved an MSE of 1.42×10^{-3}, demonstrating that the model captures underlying physical relationships rather than memorizing seasonal patterns.

We further analyzed the sensitivity of the proposed model to the hyperparameters λ_1 and λ_2, which balance the contributions of the energy balance constraint and the thermal evolution constraint, respectively. Table 6 reports the MSE as a function of λ_1 with λ_2 fixed at 1.0, and as a function of λ_2 with λ_1 fixed at 0.1. The model remains stable for λ_1 in the range of 0.05 to 0.5 and for λ_2 in the range of 0.5 to 2.0. Performance degrades when λ_1 is too small because the energy constraint becomes ineffective or too large because it dominates data loss. Similarly, excessive λ_2 forces the model to overemphasize thermal smoothness, reducing responsiveness to actual temperature variations. The current values of $\lambda_1 = 0.1$ and $\lambda_2 = 1.0$, selected via grid search on the validation set, are within the stable region and yield near-optimal performance.

Table 6. Sensitivity analysis of loss weights (PDPG, MSE $\times 10^{-3}$).

λ_1	MSE	λ_2	MSE
0.01	1.24	0.1	1.42
0.05	0.92	0.5	0.96
0.10	0.81	1.0	0.81
0.50	0.88	2.0	0.85
1.00	1.15	5.0	1.38

From the above results, three main observations can be drawn. First, PDPG consistently outperforms UEPG, suggesting that when the underlying physical relationship is already well-defined by a deterministic formula, embedding this knowledge as a fixed aggregation mechanism is more effective than learning it from data. Second, the performance gap between the full model and the variants without physics constraints indicates that the proposed PINN-GNN framework benefits from the synergy between spatial dependency modeling and physical consistency enforcement, rather than from either component alone. Third, the model's competitive performance under cross-site and cross-time settings confirms that the physics-informed constraints help capture transferable physical principles instead of memorizing environment-specific patterns. These findings collectively demonstrate the effectiveness and validity of the proposed PINN-GNN framework, suggesting that incorporating domain knowledge into neural network design offers a practical pathway toward accurate, physically consistent, and generalizable energy efficiency prediction.

5. Conclusions

In this paper, we address energy efficiency prediction as a fundamental computational method for network optimization. The predicted PUE is intended to serve as input to downstream decision-making modules. To overcome the limitations of existing approaches

in modeling spatial topology and physical constraints, we propose a general PINN-GNN hybrid framework. Data centers are used as a representative validation scenario, while the method is generalizable to various network systems. The proposed architecture integrates energy balance and thermal evolution constraints into the GNN learning process, and two schemes are developed: PDPG, which predicts device-level power consumption before computing PUE, and UEPG, which directly maps input features to PUE in an end-to-end manner. Experiments show that our models outperform mainstream baselines in accuracy and robustness. The high-precision PUE predictions provide a reliable computational foundation for network-oriented energy management tasks. As an illustrative example of how such predictions could be used in practice, when the predicted PUE exceeds a predefined threshold, an operator or an automated controller can proactively adjust cooling setpoints or redistribute workload to bring the PUE back to an efficient range.

In future work, we will extend the proposed prediction method to other network scenarios beyond data centers, including cloud-edge computing infrastructures and wireless communication networks. We will also explore integrating the prediction framework with intelligent scheduling strategies towards closed-loop optimization for network energy efficiency, contributing to the development of green and low-carbon 6G networks.

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